Python基础综合案例

数据可视化 - 地图可视化

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- 基础地图使用
- 疫情地图-国内疫情地图
- 疫情地图-省级疫情地图

学习目标

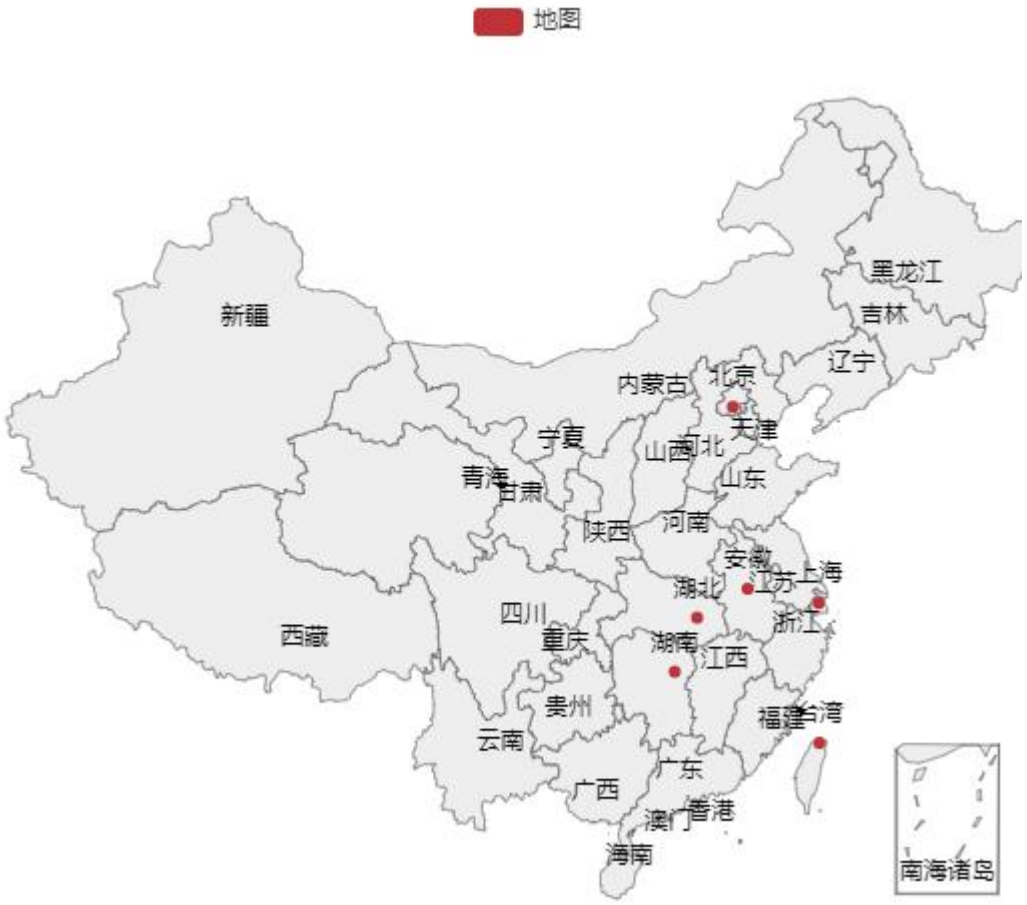
Learning Objectives

1. 掌握使用pyecharts构建基础的全国地图可视化图表

基础地图演示

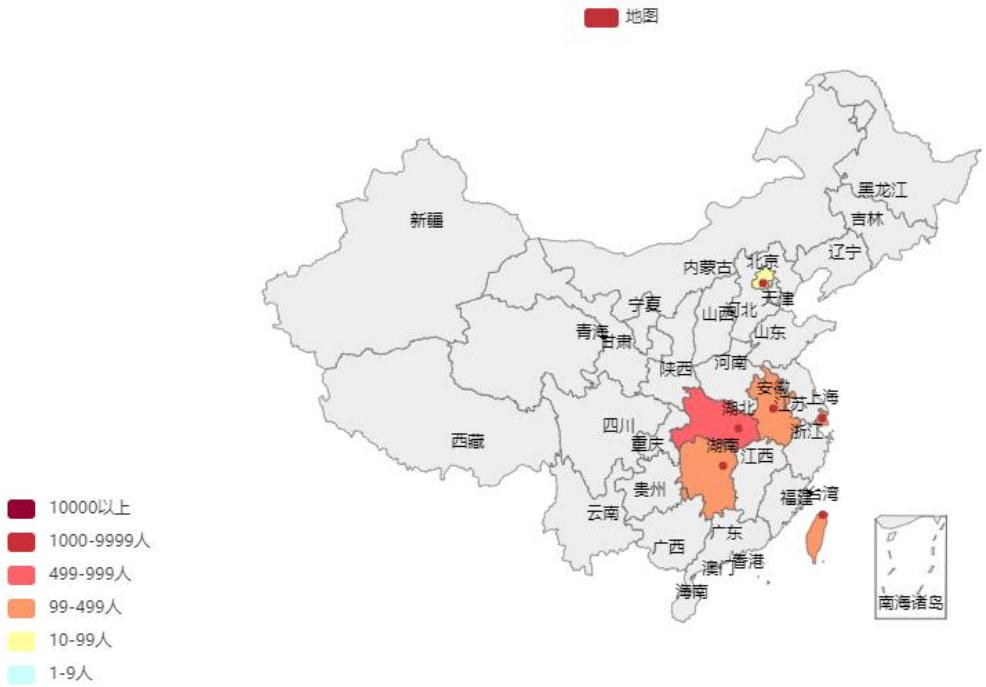
```
from pyecharts.charts import Map
from pyecharts.options import VisualMapOpts

map = Map()
data = [
    ("北京", 99),
    ("上海", 199),
    ("湖南", 299),
    ("台湾", 199),
    ("安徽", 299),
    ("广州", 399),
    ("湖北", 599)
]
map.add("地图", data, "china")
```



基础地图演示 – 视觉映射器

```
map.set_global_opts(
    visualmap_opts=VisualMapOpts(
    is_show=True,
    is_piecewise=True,
    pieces=[
        {"min": 1, "max": 9, "label": "1-9人", "color": "#CCFFFF"},
        {"min": 10, "max": 99, "label": "10-99人", "color": "#FFFF99"},
        {"min": 100, "max": 499, "label": "99-499人", "color": "#FF9966"},
        {"min": 500, "max": 999, "label": "499-999人", "color": "#FF6666"},
        {"min": 1000, "max": 9999, "label": "1000-9999人", "color": "#CC3333"},
        {"min": 10000, "label": "10000以上", "color": "#990033"}
    ])
)
```



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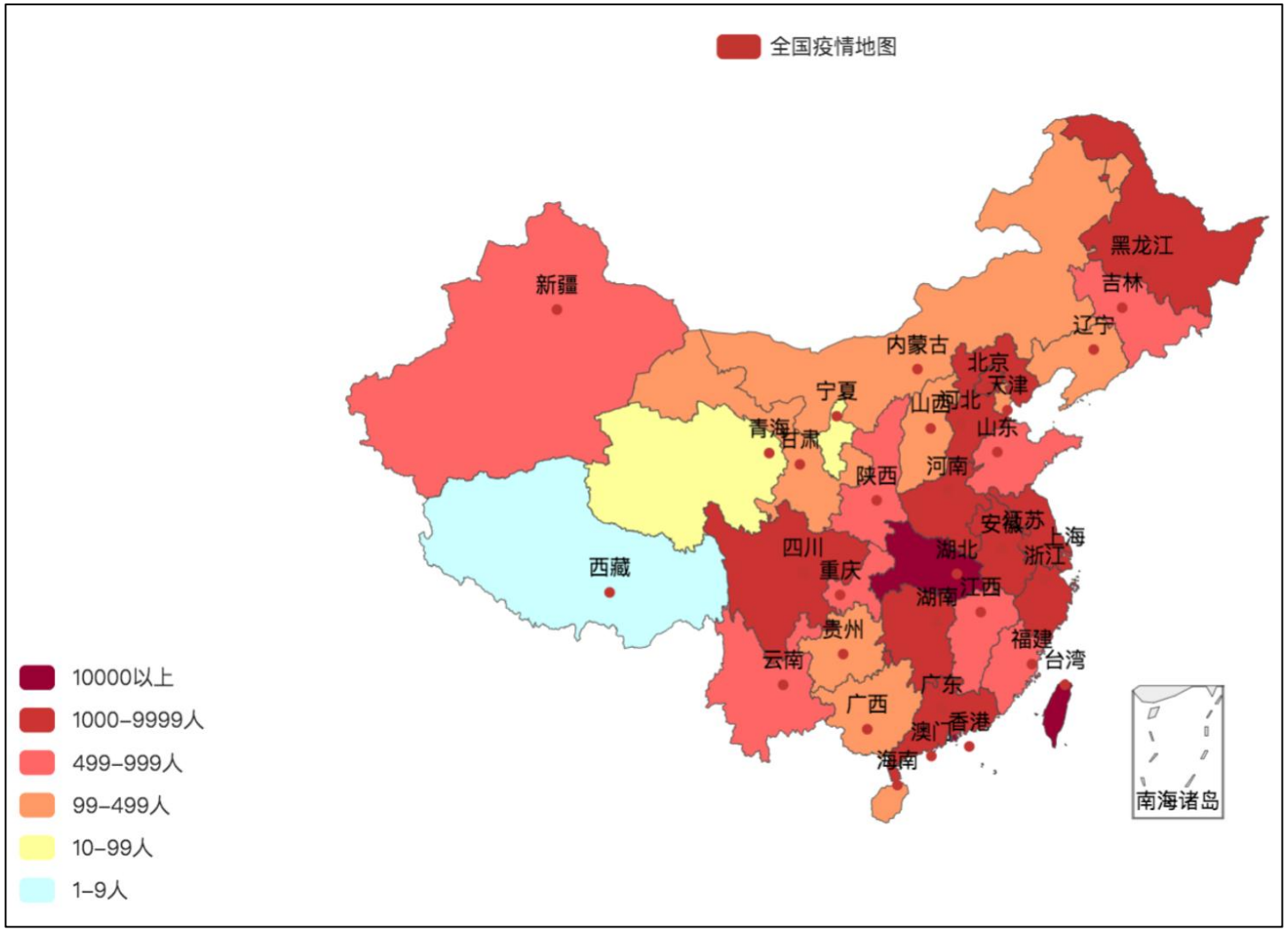
案例效果


全国疫情地图
新疆
西藏
黑龙江
吉林
辽宁
内蒙古
北京
天津
宁夏
山西
河北
山东
青海
甘肃
陕西
河南
安徽
苏
上海
四川
湖北
浙江
重庆
湖南
江西
贵州
福建
台湾
云南
广东
广西
澳门
香港
海南
南海诸岛
10000以上
1000-9999人
499-999人
99-499人
10-99人
1-9人

数据整理

- 获取数据

```
# 导入json模块
import json

# 获取数据
with open("疫情.txt") as f:
    data = f.read()

# 对数据进行处理
data = json.loads(data)
```

数据整理

- 数据整体结构（全国）


```
{
    "lastUpdateTime":"2021-08-18 10:53:30",
    "chinaTotal":
        {"confirm":122482,"heal":112984,"dead":5669,"nowConfirm":3829,"suspect":1,"nowSevere":62,"importedCase
    "chinaAdd":
        {"confirm":49,"heal":91,"dead":0,"nowConfirm":-42,"suspect":0,"nowSevere":-5,"importedCase":22,"noInfe
    "isShowAdd":true,
    "showAddSwitch":
        {"all":true,"confirm":true,"suspect":true,"dead":true,"heal":true,"nowConfirm":true,"nowSevere":true,"
    "areaTree": 地区
        [
            {
            "name":"中国", 国家
            "today":{"confirm":49,"isUpdated":true},
            "total":{"nowConfirm":3829,"confirm":122482,"suspect":1,"dead":5669,"deadRate":"4.63","showRate":f
            "children":
                [
                    {"name":"台湾",  省
                        "today":{"confirm":18,"confirmCuts":0,"isUpdated":true,"tip":"","wzz_add":0},
                        "total":{"nowConfirm":1864,"confirm":15880,"suspect":0,"dead":821,"deadRate":"5.17","s
                        "children":
                        [
                            {
                            "name":"地区待确认",
                            "today":{"confirm":18,"confirmCuts":0,"isUpdated":true},
                            "total":{"nowConfirm":1864,"confirm":15880,"suspect":0,"dead":821,"deadRate":"5.17
                            }
                        ]
                    },

                    {"name":"江苏",  省
                        "today":{"confirm":6,"confirmCuts":0,"isUpdated":true,"tip":"","wzz_add":0},
                        "total":{"nowConfirm":711,"confirm":1576,"suspect":0,"dead":0,"deadRate":"0.00","showR
                        "children":
                        [
                            {
```

数据整理

- 省数据结构


```
{"name":"江苏",  省
    "today":{"confirm":6,"confirmCuts":0,"isUpdated":true,"tip":"","wzz_add":0},
    "total":{"nowConfirm":711,"confirm":1576,"suspect":0,"dead":0,"deadRate":"0.00","showF
    "children":                          确诊总数
    [
        {
        "name":"扬州",  市
        "today":{"confirm":6,"confirmCuts":0,"isUpdated":true},
        "total":{"nowConfirm":541,"confirm":584,"suspect":0,"dead":0,"deadRate":"0.00","sh
        {"name":"南京",
        "today":{"confirm":0,"confirmCuts":0,"isUpdated":true},
        "total":{"nowConfirm":160,"confirm":329,"suspect":0,"dead":0,"deadRate":"0.00","sh
        {"name":"境外输入",
        "today":{"confirm":0,"confirmCuts":0,"isUpdated":true},
        "total":{"nowConfirm":13,"confirm":133,"suspect":0,"dead":0,"deadRate":"0.00","sho
        {"name":"淮安",
        "today":{"confirm":0,"confirmCuts":0,"isUpdated":false},
        "total":{"nowConfirm":12,"confirm":78,"suspect":0,"dead":0,"deadRate":"0.00","show
        {"name":"宿迁",
```

数据整理

- 获取每个省份的确诊数据：

```
f = open("D:/疫情.txt", "r", encoding="UTF-8")
data = json.loads(f.read())

# data["areaTree"][0]["children"]为各省数据
data = data["areaTree"][0]["children"]

# 创建一个空列表存储相关数据
data_list = []

# 循环data的元素，将省和数据封装到元组中，每一个元组作为列表的一个元素
for province_data in data:
    province_name = province_data["name"]                    # 省名称
    province_value = province_data["total"]["confirm"]       # 省确诊

    # 添加到数据list内
    data_list.append((province_name, province_value))

# 输出数据
print(data_list)
```

数据整理

- 上述代码执行后输出，每个省的确诊数据：

```
[('台湾', 15880), ('江苏', 1576), ('云南', 982), ('河南', 1518), ('上海', 2408), ('湖南', 1181),
```

国内疫情地图

- 创建地图并添加数据
  - 导入模块：

```
from pyecharts.charts import Map
```

  - 创建地图：

```
# 这里运行后不会显示任何结果 需要给地图添加数据才会有相应的结果
virus_map = Map()
```

国内疫情地图

- 添加数据：

```
# 各省名称
print(result_dict.keys())
# 各省确诊人数
print(result_dict.values())
```

```
dict_keys(['台湾', '江苏', '云南', '河南', '上海',
dict_values([15880, 1576, 982, 1518, 2408, 1181,
```

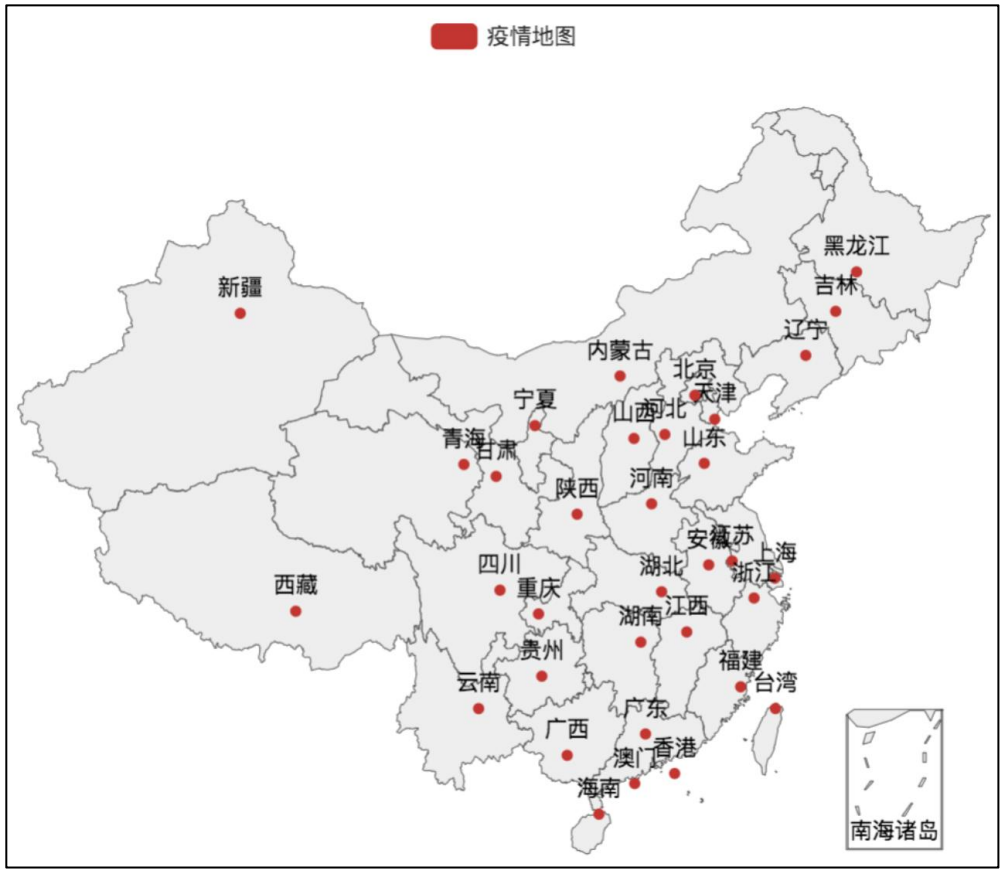
```
# 省名称和确诊人数对应列表
data_list = zip(result_dict.keys(),result_dict.values())
data_list = list(data_list)
print(data_list)
```

```
[('台湾', 15880), ('江苏', 1576), ('云南', 982), ('河南', 1518), ('上海', 2408),
```

国内疫情地图

- 添加数据：

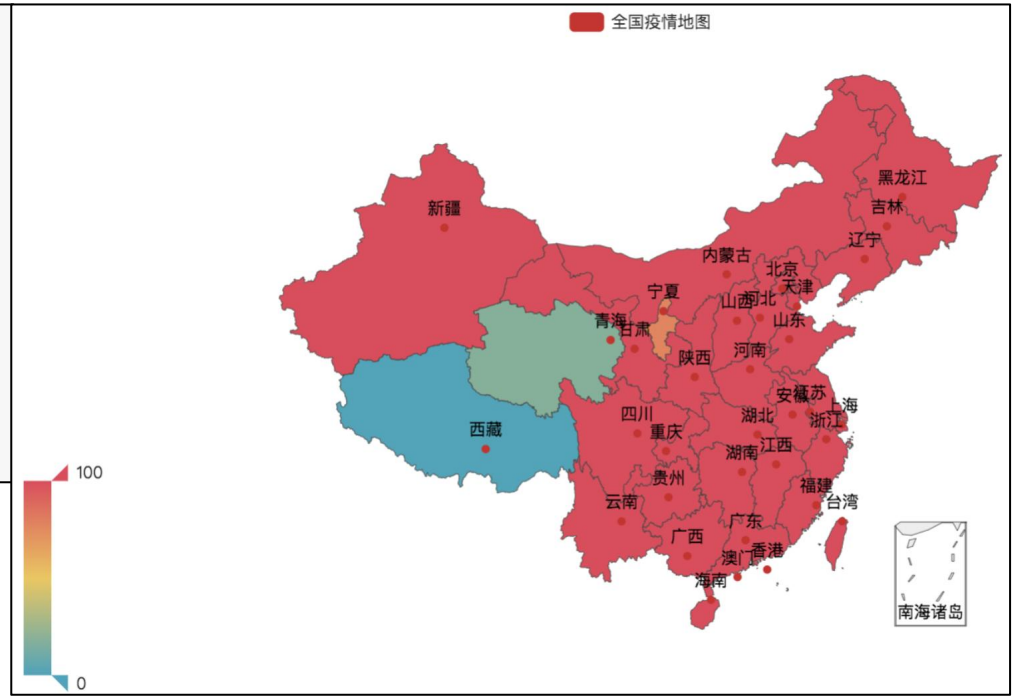
```
# add函数 ： 给地图添加数据
# 参数1: 解释标签
# 参数2: 数据(列表类型)
# 参数3: 地图所属位置
virus_map = virus_map.add("疫情地图", data_list, "china")
# 生成结果
virus_map.render()
```



国内疫情地图

- 设置全局配置选项

```
# 导入配置选项模块
from pyecharts import options as opts
# set_global_opts()函数 : 设置全局配置选项
# visualmap_opts=opts.VisualMapOpts() : 视觉映射配置选项
virus_map.set_global_opts(visualmap_opts=opts.VisualMapOpts())
# 生成结果
virus_map.render()
```



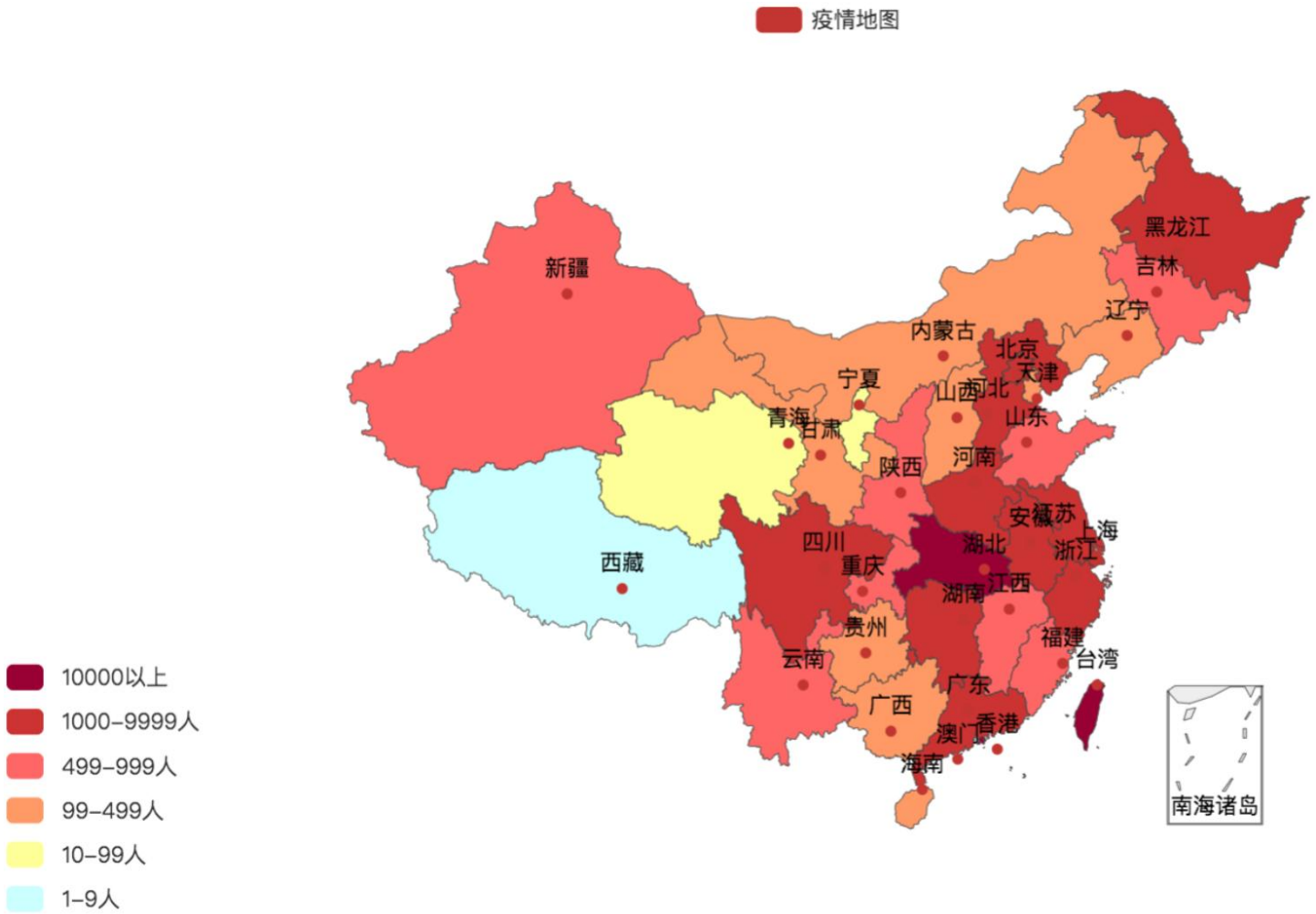
国内疫情地图

- 设置全局配置选项

```
# is_piecewise : 是否为分段型
# pieces : 自定义的每一段的范围，以及每一段的文字，以及每一段的特别的样式
virus_map = virus_map.set_global_opts(visualmap_opts=opts.VisualMapOpts(
                                                    is_piecewise=True,
                                                    pieces=[
                                                                {"min": 1, "max": 9, "label": "1-9人", "color": "#CCFFFF"},
                                                                {"min": 10, "max": 99, "label": "10-99人", "color": "#FFFF99"},
                                                                {"min": 100, "max": 499, "label": "99-499人", "color": "#FF9966"},
                                                                {"min": 500, "max": 999, "label": "499-999人", "color": "#FF6666"},
                                                                {"min": 1000, "max": 9999, "label": "1000-9999人", "color": "#CC3333"},
                                                                {"min": 10000, "label": "10000以上", "color": "#990033"}
                                                                  ] ))
```

国内疫情地图

- 设置全局配置选项



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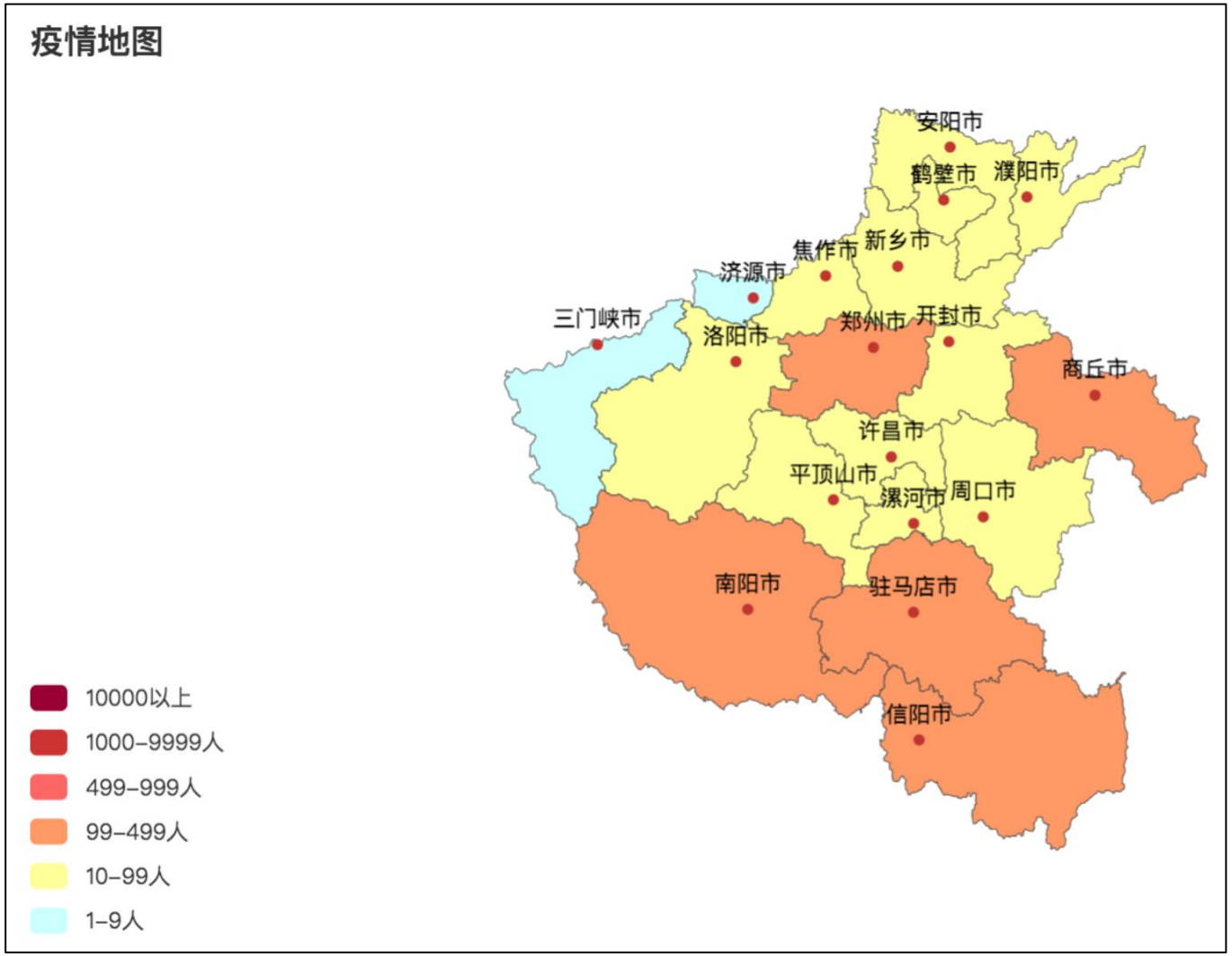
- 疫情地图-省级疫情地图

省疫情地图

- 效果展示


疫情地图
安阳市
鹤壁市
濮阳市
焦作市
新乡市
济源市
三门峡市
洛阳市
郑州市
开封市
商丘市
许昌市
平顶山市
漯河市
周口市
南阳市
驻马店市
信阳市
10000以上
1000-9999人
499-999人
99-499人
10-99人
1-9人

省疫情地图

- 获取河南省各市数据

```
# 导入模块
import json
from pyecharts.charts import Map
from pyecharts import options as opts
# 获取全国疫情数据
with open("疫情.txt") as f:
    data = f.read()
# 对数据进行转化
data = json.loads(data)
# 获取各省确诊数据
data = data["areaTree"][0]["children"]
# 对数据过滤，只留下河南省的各市区的确诊数据
for i in data:
    if i["name"] == "河南":
        data = i["children"]
```

省疫情地图

- 省数据结构：


```
{"name":"江苏",  省
    "today":{"confirm":6,"confirmCuts":0,"isUpdated":true,"tip":"","wzz_add":0},
    "total":{"nowConfirm":711,"confirm":1576,"suspect":0,"dead":0,"deadRate":"0.00","showP
    "children":                    确诊总数
    [
        {
        "name":"扬州",  市
        "today":{"confirm":6,"confirmCuts":0,"isUpdated":true},
        "total":{"nowConfirm":541,"confirm":584,"suspect":0,"dead":0,"deadRate":"0.00","sh
        {"name":"南京",
        "today":{"confirm":0,"confirmCuts":0,"isUpdated":true},
        "total":{"nowConfirm":160,"confirm":329,"suspect":0,"dead":0,"deadRate":"0.00","sh
        {"name":"境外输入",
        "today":{"confirm":0,"confirmCuts":0,"isUpdated":true},
        "total":{"nowConfirm":13,"confirm":133,"suspect":0,"dead":0,"deadRate":"0.00","sho
        {"name":"淮安",
        "today":{"confirm":0,"confirmCuts":0,"isUpdated":false},
        "total":{"nowConfirm":12,"confirm":78,"suspect":0,"dead":0,"deadRate":"0.00","show
        {"name":"宿迁",
```

省疫情地图

- 把各市数据汇总到一个列表中：

```
f = open("D:/疫情.txt", "r", encoding="UTF-8")
data = json.loads(f.read())

# data["areaTree"][0]["children"]为各省数据
data = data["areaTree"][0]["children"]

# 创建一个空列表存储相关数据
data_list = []

# data[3]是河南数据
for city_data in data[3]['children']:
    data_list.append(
      (city_data["name"] + "市", city_data["total"]["confirm"])
    )
data_list.append(("济源市", 5))    # 济源市没有数据，手动补上
# 输出数据
print(data_list)
```

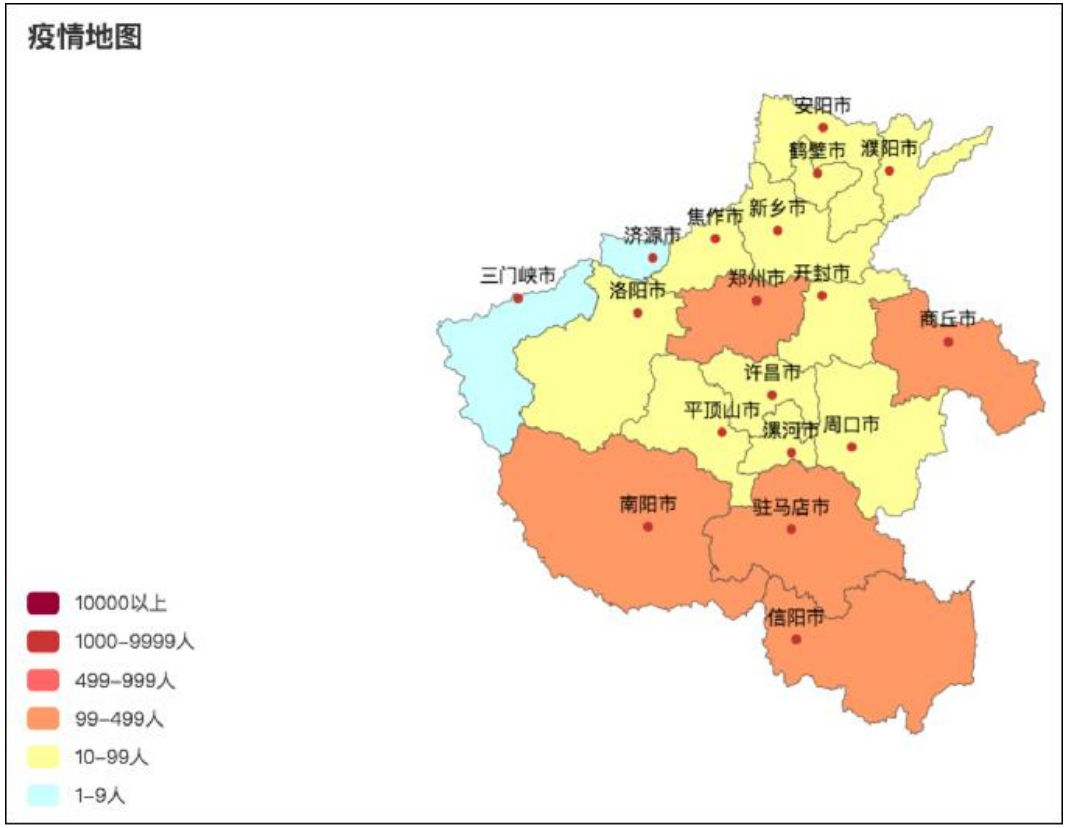
```
[('郑州', 295), ('境外输入', 79), ('商丘', 106), ('开封', 33), ('驻马店', 143), ('安阳', 54), ('许昌', 40),
```

省疫情地图

- 参考国内疫情地图生成河南省疫情地图：

```
map = Map()
map.add("河南省疫情数据", data_list, "河南")
map.set_global_opts(
    visualmap_opts=VisualMapOpts(
        is_show=True,
        is_piecewise=True,
        pieces=[
            {"min": 1, "max": 9, "label": "1-9人", "color": "#CCFFFF"},
            {"min": 10, "max": 99, "label": "10-99人", "color": "#FFFF99"},
            {"min": 100, "max": 499, "label": "99-499人", "color": "#FF9966"},
            {"min": 500, "max": 999, "label": "499-999人", "color": "#FF6666"},
            {"min": 1000, "max": 9999, "label": "1000-9999人", "color": "#CC3333"},
            {"min": 10000, "label": "10000以上", "color": "#990033"}
        ]
    )
)

map.render("河南省.html")
```



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